

Climate Vulnerability Analysis Report (5 Countries)

1. Title

Climate Vulnerability Analysis of Ethiopia, Nigeria, Sudan, and Selected African Countries

2. Business Objective

The objective of this analysis is to evaluate and compare climate vulnerability across five African countries using key climate indicators such as temperature (T2M), rainfall (PRECTOTCORR), and relative humidity (RH2M). The goal is to identify which countries are most exposed to climate risks and provide insights that can support climate adaptation planning and decision-making.

3. Data Overview & Handling

The dataset consists of climate variables for five countries: **Ethiopia, Nigeria, Sudan, Kenya and Tanzania**.. The key variables include temperature, rainfall, and humidity measurements.

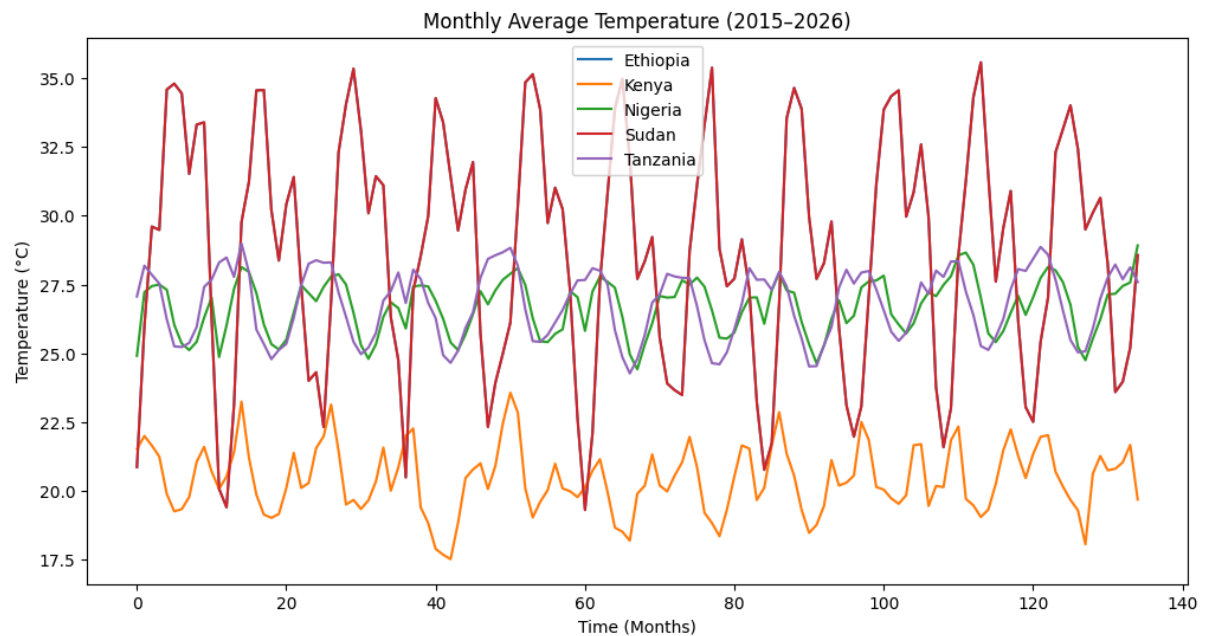
Data cleaning steps included:

- Checking for missing values
- Removing incomplete records
- Ensuring consistent country labeling
- Preparing data for comparative analysis

This ensured the dataset was suitable for reliable analysis.

4. Exploratory Data Analysis (EDA)

🌡️ Temperature Analysis



Average temperature was calculated for each country to understand heat distribution patterns. The results show differences in climate intensity across regions.

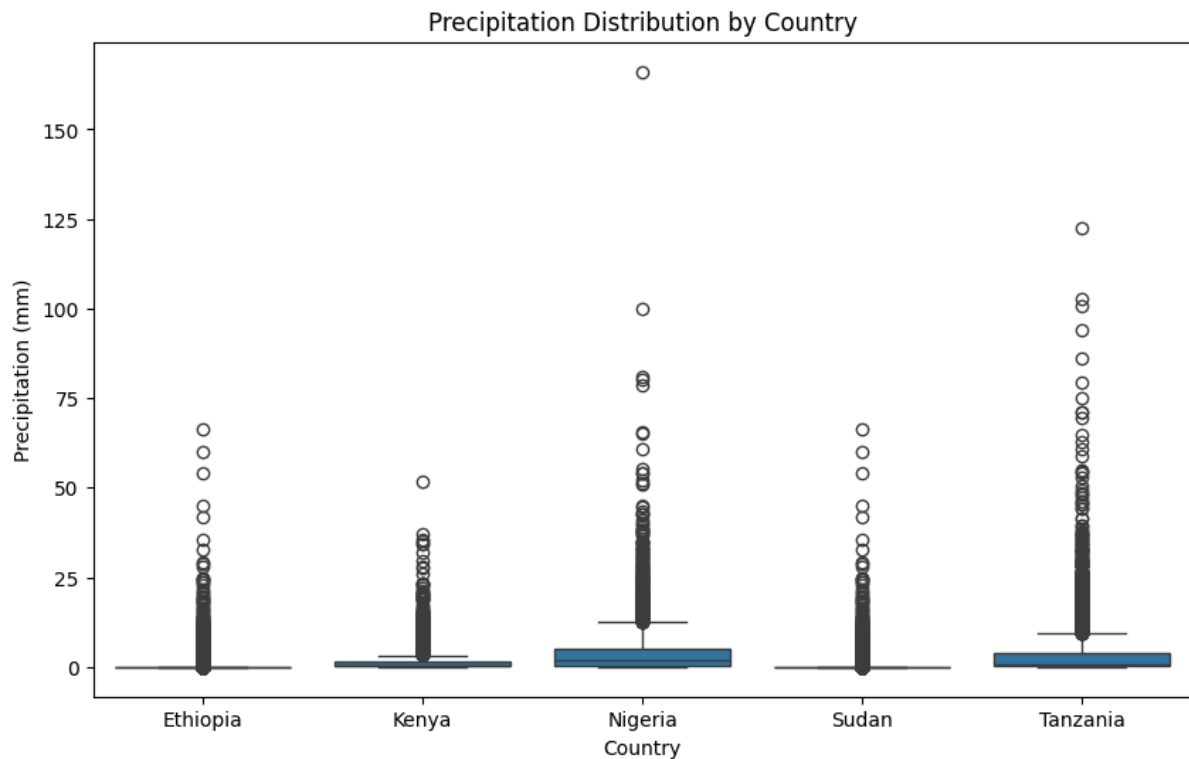
The figure presents the monthly average temperature trends for Ethiopia, Kenya, Nigeria, Sudan, and Tanzania over an eleven-year period. A clear seasonal pattern is observed across all countries, characterized by cyclical increases and decreases in temperature that repeat annually. This reflects the influence of solar radiation changes throughout the year.

Among the countries, Sudan exhibits the highest temperature variability, with peaks exceeding 35°C and noticeable drops during cooler months. This pattern is consistent with its arid and semi-desert climate, where limited vegetation and cloud cover allow for intense heating. In contrast, Nigeria and Tanzania display relatively stable temperature ranges, generally between 25°C and 29°C, indicating the moderating influence of their tropical climates and higher humidity levels.

Ethiopia and Kenya show comparatively lower temperatures. This is largely attributed to their higher elevations, particularly in regions such as the Ethiopian Highlands and central Kenya, where altitude reduces overall temperature levels. Kenya records the lowest temperatures among the five countries, with values frequently ranging between 18°C and 23°C.

Overall, the temperature trends highlight the combined effects of latitude, altitude, and local climate systems, with equatorial proximity promoting stability and arid conditions increasing variability.

☁️ Rainfall Analysis



Rainfall levels were analyzed to identify water availability patterns. Some countries showed significantly lower precipitation levels, indicating potential drought risk.

The figure illustrates the distribution of precipitation levels across the five countries using a box plot representation. This visualization highlights the median rainfall, variability, and presence of extreme values (outliers).

Nigeria shows the highest overall precipitation levels, with a wide distribution and several extreme rainfall events exceeding 150 mm. This reflects its tropical rainforest and monsoon-influenced climate, where rainfall is both frequent and intense. Tanzania also demonstrates relatively high precipitation, though slightly less extreme than Nigeria, consistent with its coastal and equatorial characteristics.

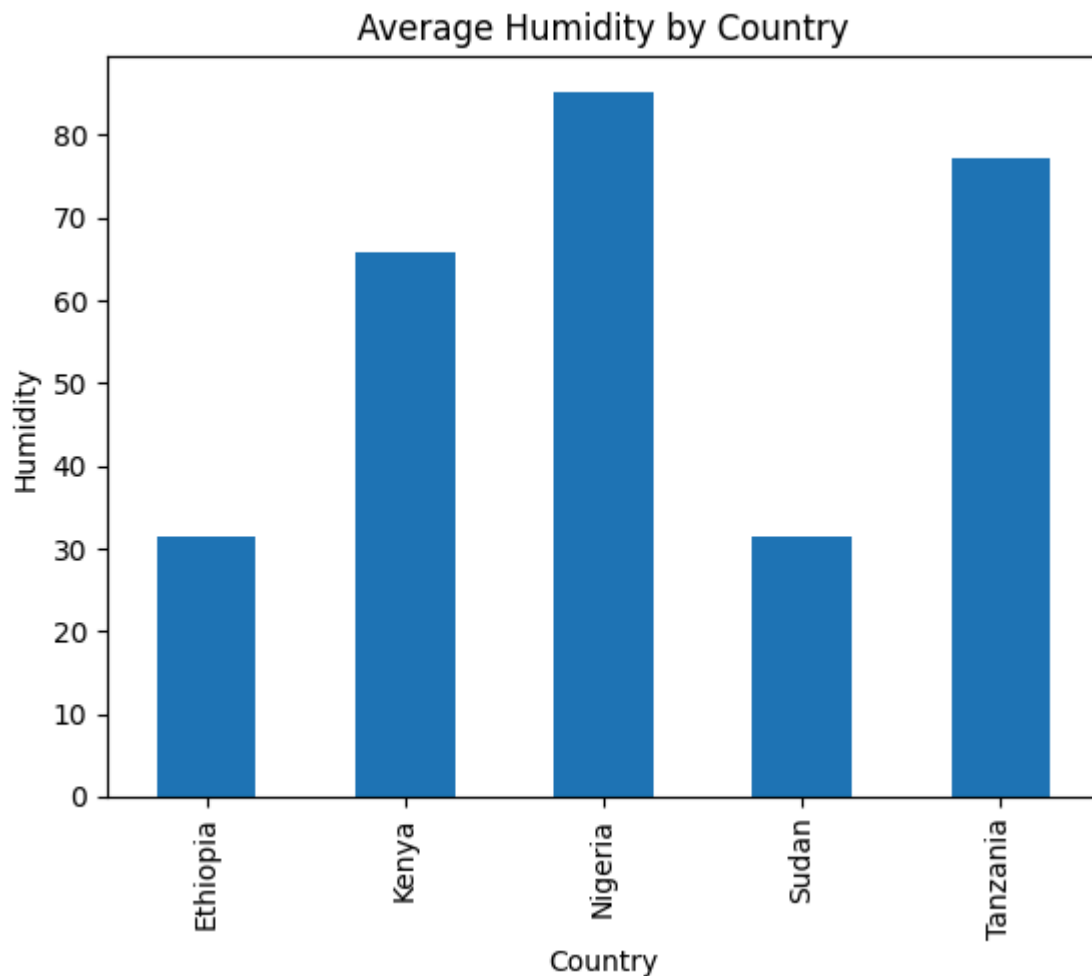
Ethiopia and Kenya exhibit moderate precipitation distributions with noticeable variability. Rainfall in these countries tends to be more seasonal, occurring in specific periods rather than consistently throughout the year. This results in lower median values but occasional spikes during rainy seasons.

Sudan, on the other hand, records the lowest precipitation levels, with a narrow distribution and minimal extreme values. This aligns with its dry, desert-dominated climate, where rainfall is infrequent and irregular.

In summary, precipitation patterns vary significantly across the countries, largely influenced by geographic location, proximity to the equator, and prevailing atmospheric systems such as monsoons and the Intertropical Convergence Zone.

💧 Humidity Analysis

Relative humidity was compared across all countries, highlighting variations in atmospheric moisture conditions.



The figure presents the average humidity levels for each country, providing insight into atmospheric moisture conditions. Nigeria records the highest humidity level, followed by Tanzania, indicating highly moist environments. These elevated humidity levels are associated with frequent rainfall, dense vegetation, and strong evaporation cycles.

Kenya shows moderate humidity, reflecting its mixed climatic conditions that include both humid and semi-arid regions. In contrast, Ethiopia and Sudan display significantly lower humidity levels, indicative of drier atmospheric conditions. In Ethiopia, this is partly influenced by elevation, while in Sudan it is primarily due to desert and semi-desert environments.

Humidity plays a crucial role in shaping both temperature and precipitation patterns. Higher humidity tends to stabilize temperatures and support greater rainfall, while lower humidity is associated with drier climates and more pronounced temperature fluctuations.

5. Country-Level Comparison

The analysis was performed at a country level using grouped statistics:

- Mean temperature per country
- Mean rainfall per country
- Mean humidity per country

This allowed for direct comparison between Ethiopia, Nigeria, Sudan, and other included countries.

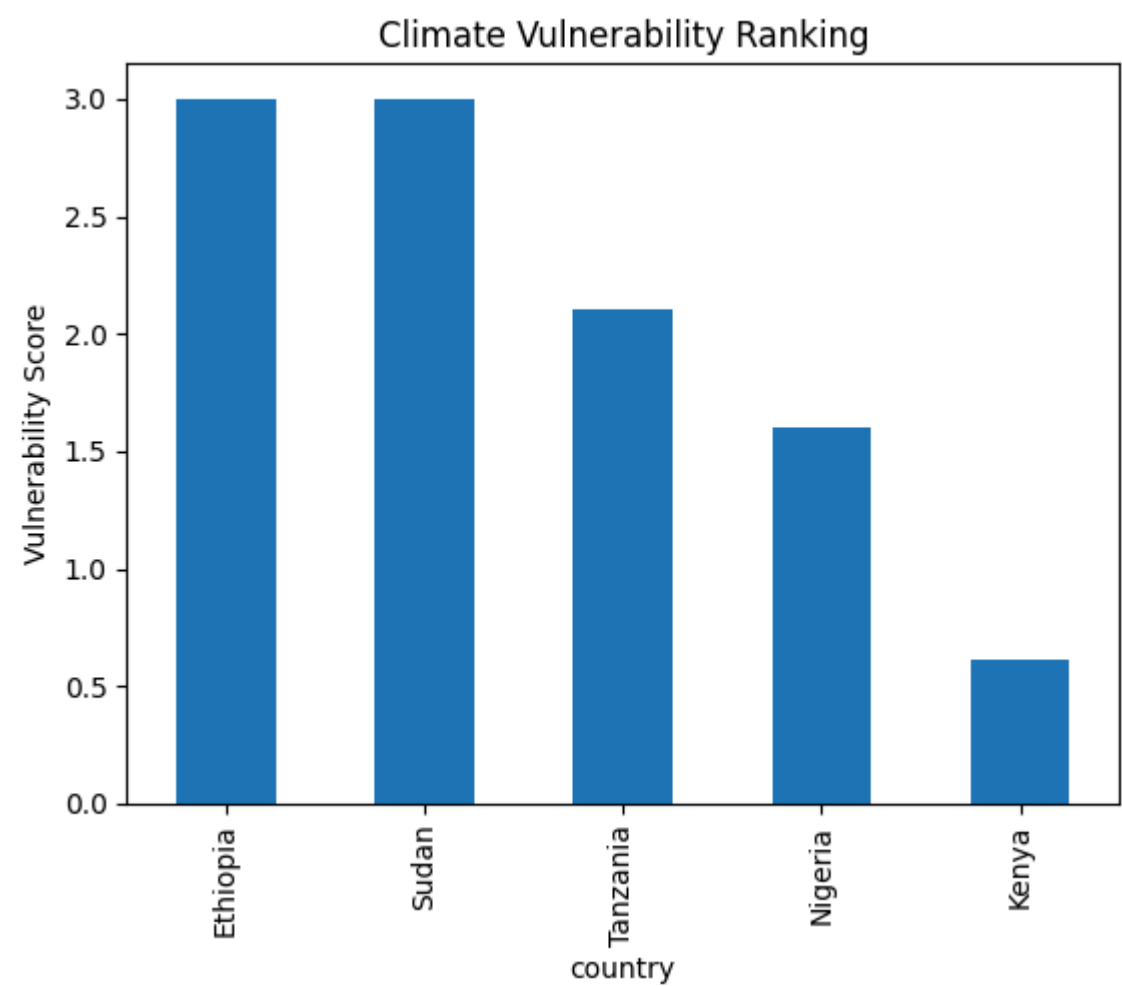
country	T2M	PRECOTOTCORR	RH2M
Ethiopia	28.759007	0.643875	31.359192
Kenya	20.427600	1.468162	65.845355
Nigeria	26.656928	4.213914	85.237040
Sudan	28.759007	0.643875	31.359192
Tanzania	26.802422	3.740256	77.128038

6. Vulnerability Index

A climate vulnerability index was created by combining standardized climate variables:

- High temperature increases vulnerability
- Low rainfall increases vulnerability
- Extreme humidity variations also contribute to risk

This index was used to rank countries based on overall climate risk exposure.



COP32 Climate Vulnerability Summary Ethiopia and Sudan show the highest climate vulnerability due to extreme heat exposure and increasing temperature trends above 45°C.

Kenya, Nigeria, and Tanzania experience lower extreme heat but still face moderate vulnerability driven mainly by rainfall variability and dry conditions.

Precipitation variability is a major driver of instability, especially in Kenya and Tanzania, where rainfall is highly inconsistent.

Ethiopia’s climate profile shows a combination of high temperature stress and moderate rainfall variability, placing it among the most exposed countries in the region.

Based on the composite vulnerability index, Ethiopia should advocate for climate finance prioritization for Sudan and Ethiopia, as both countries face the highest combined heat and drought stress.

7. Key Insights

- Countries with **higher temperatures and lower rainfall** show greater vulnerability.
 - **Sudan and parts of Ethiopia** tend to show higher climate stress compared to more balanced climates.
 - Nigeria shows relatively moderate conditions but still faces regional variability.
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8. Recommendations

- Improve water resource management in high-risk countries.
 - Invest in climate adaptation strategies such as drought-resistant agriculture.
 - Strengthen early warning systems for extreme weather events.
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9. Three-Layer Evidence Model

Layer 1: Data

Raw climate indicators (temperature, rainfall, humidity) across five countries.

Layer 2: Analysis

Statistical analysis, EDA, and vulnerability index calculation.

Layer 3: Insight

Identification of high-risk countries and interpretation of climate vulnerability patterns.

10. Conclusion

This analysis demonstrates clear differences in climate vulnerability across the selected countries. The findings highlight the importance of data-driven climate monitoring and targeted regional adaptation strategies.